

Introduction to the Practice of
STATISTICS NINTH
EDITION

Moore / McCabe / Craig

Chapter 2: Looking at Data — Relationships

Chapter 2

Looking at Data— Relationships

2.1 Relationships

2.2 Scatterplots

2.3 Correlation

2.4 Least-Squares Regression

**2.5 Cautions about Correlation and
Regression**

2.6 Data Analysis for Two-Way Tables

2.7 The Question of Causation

2.1 Relationships

- What is an association between variables?
- Explanatory and response variables
- Key characteristics of a data set

Associations Between Variables

Many interesting examples of the use of statistics involve relationships between pairs of variables.

Two variables measured on the same cases are **associated** if knowing the value of one of the variables tells you something that you would not otherwise know about the value of the other variable.

When you examine the relationship between two variables, a new question becomes important: *Is your purpose simply to explore the nature of the relationship, or do you wish to show that one of the variables can explain variation in the other?*

A **response variable** measures an outcome of a study. An **explanatory variable** may help explain or predict changes in the response variable.

Scatterplot 1

The most useful graph for displaying the relationship between two quantitative variables is a **scatterplot**.

A **scatterplot** shows the relationship between two quantitative variables measured on the same cases. The values of one variable appear on the horizontal axis, and the values of the other variable appear on the vertical axis. *Each case corresponds to one point on the graph.*

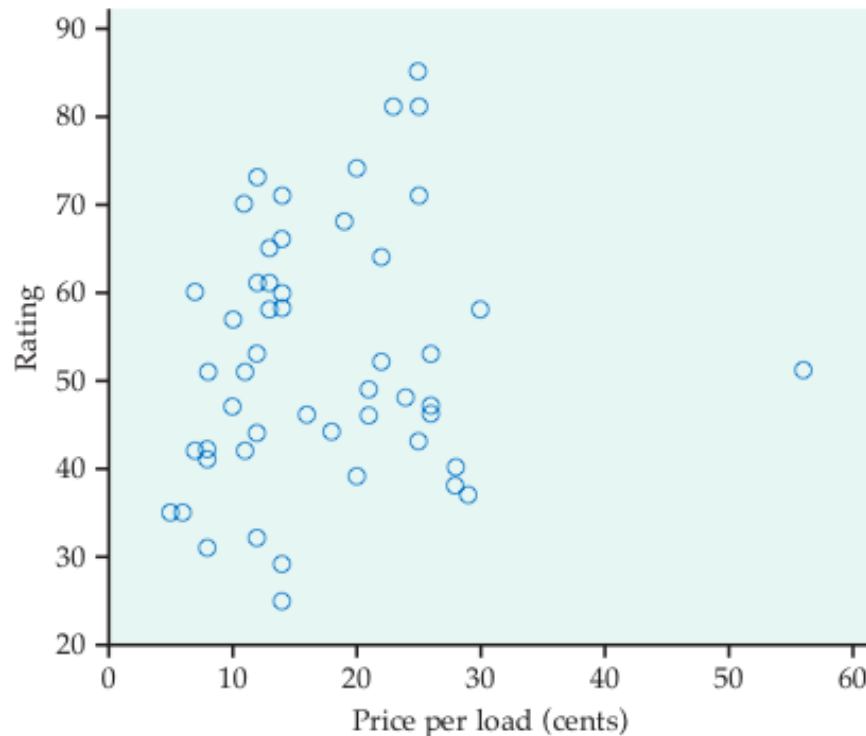
How to Make a Scatterplot

1. Decide which variable should go on each axis. If a distinction exists, *plot the explanatory variable on the x axis and the response variable on the y axis.*
2. Label and scale your axes.
3. Plot individual data values.

Scatterplot 2

Example: Make a scatterplot of the relationship between price and rating for laundry detergents.

Price per load (cents)	14	14	14	12	10	8	...	12
Rating	60	58	60	53	57	41	...	32



Interpreting Scatterplots 1

How to Examine a Scatterplot

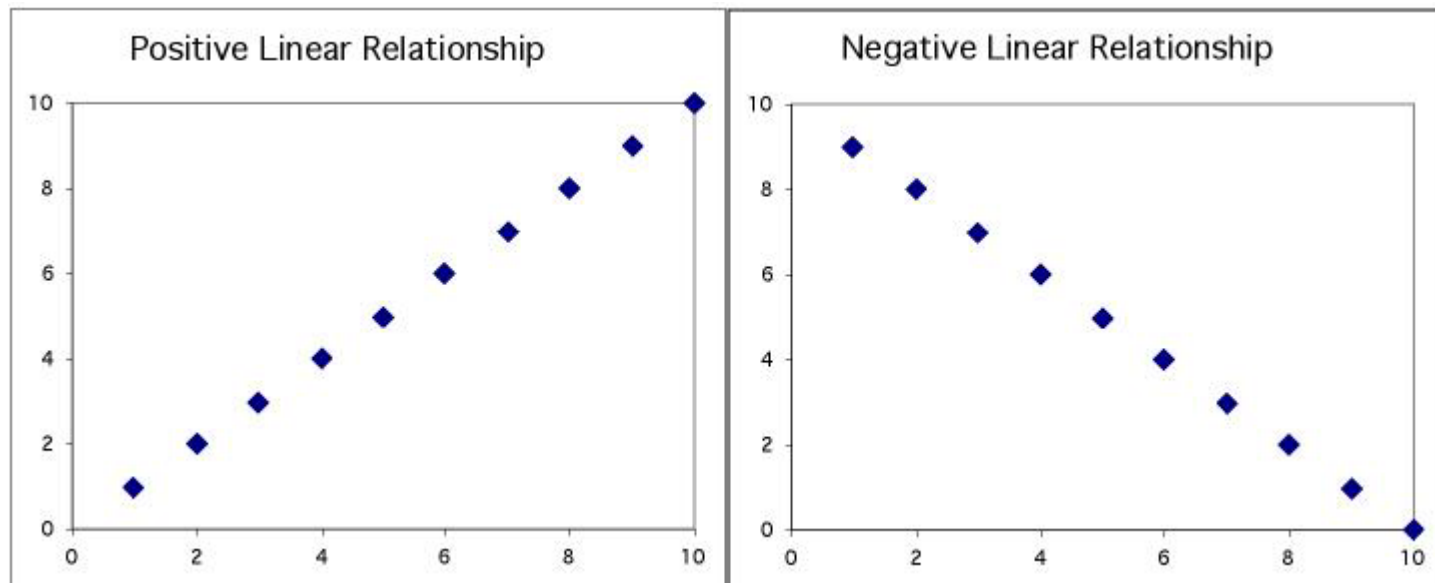
As in any graph of data, look for the *overall pattern* and for striking *deviations* from that pattern.

- You can describe the overall pattern of a scatterplot by the **form**, **direction**, and **strength** of the relationship.
- An important kind of departure is an **outlier**, an observation that falls outside the overall pattern of the relationship.

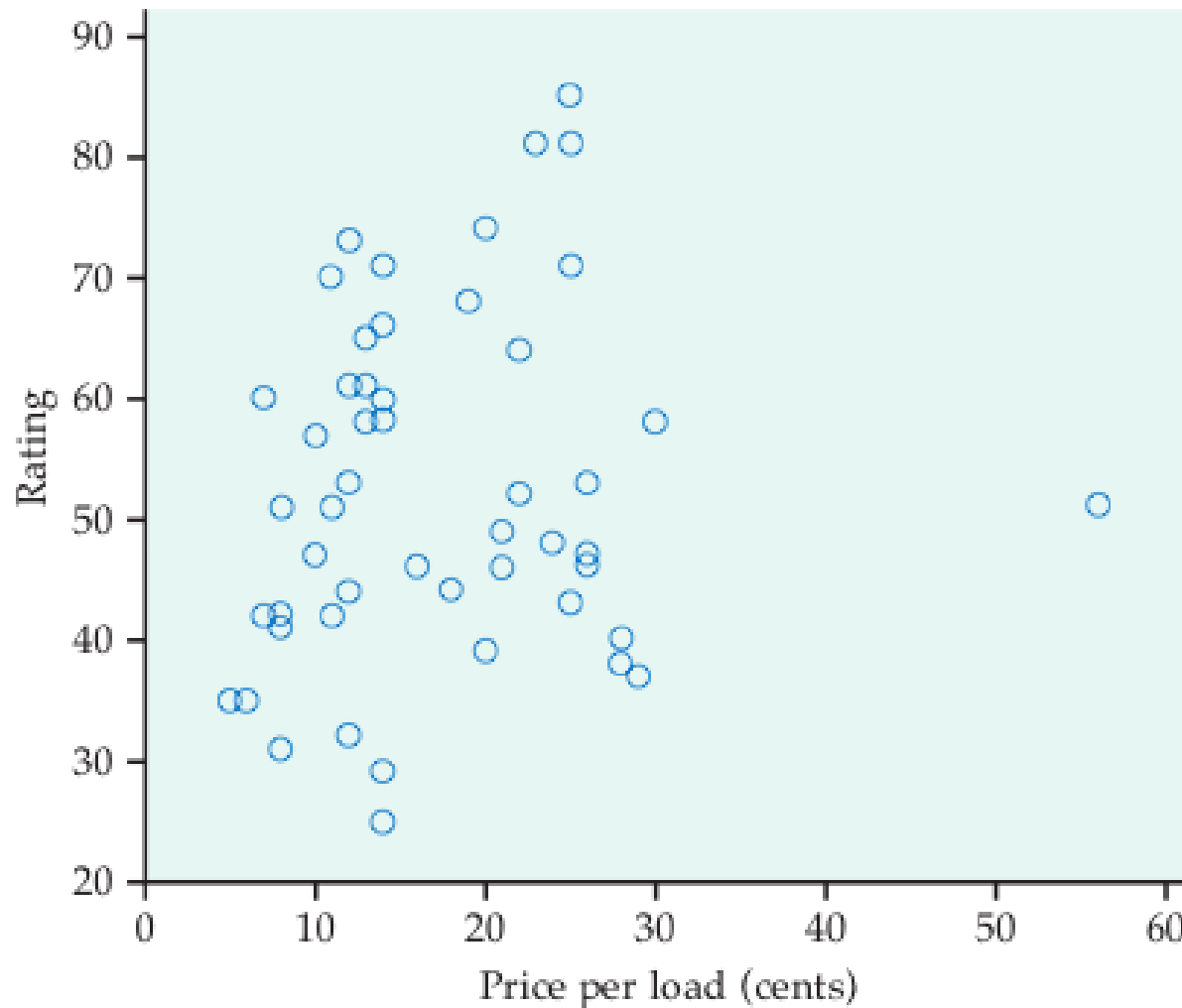
Interpreting Scatterplots 2

Two variables are **positively associated** when above-average values of one tend to accompany above-average values of the other and when below-average values also tend to occur together.

Two variables are **negatively associated** when above-average values of one tend to accompany below-average values of the other and vice versa.

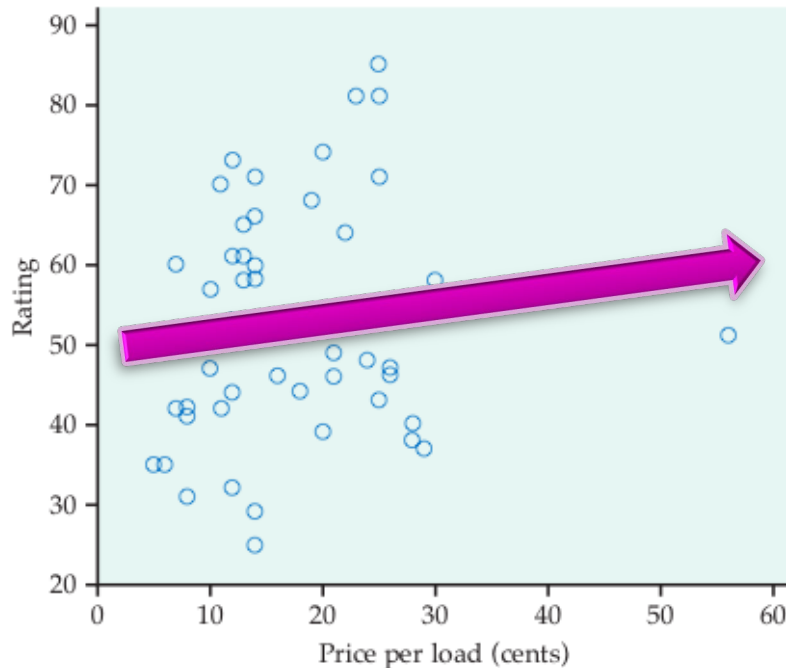


Interpreting Scatterplots 3



Interpreting Scatterplots 4

10



Outlier

- ✓ There is one possible outlier—the detergent with the price per load of 56 cents seems to have a lower rating than one might expect if higher price means a higher rated detergent.

Strength

Direction

Form

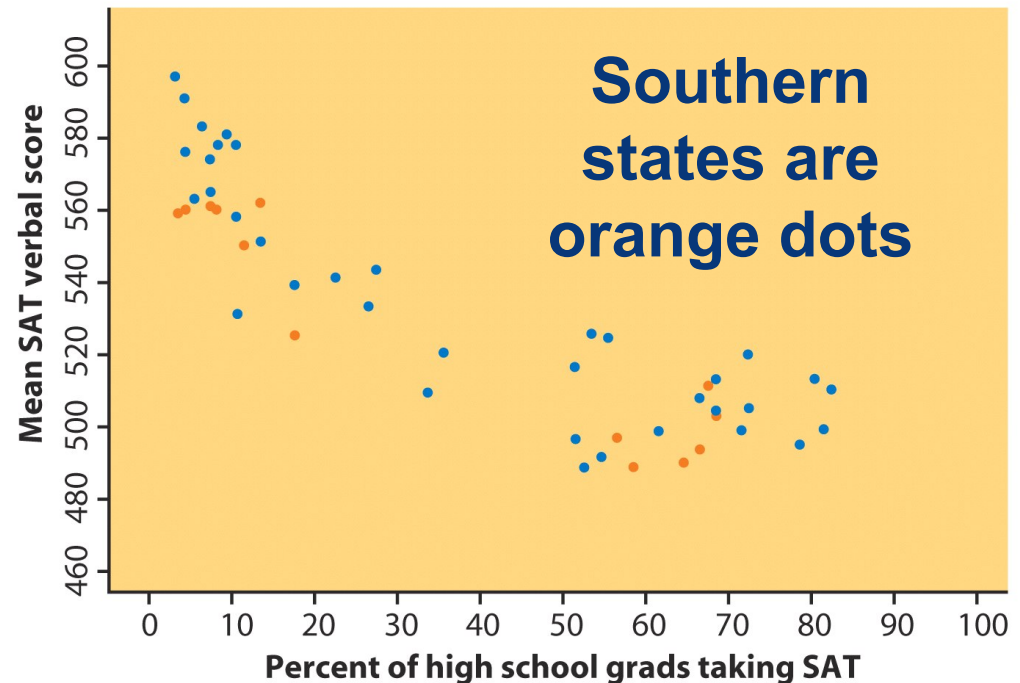
- ✓ There is a moderately weak, positive, linear relationship between price per load and detergent rating.
- ✓ It appears that more expensive detergents are rated slightly higher, with possibly one exception.

Adding Categorical Variables

10

- Consider the relationship between mean SAT verbal score and percent of high school grads taking the SAT for each state.

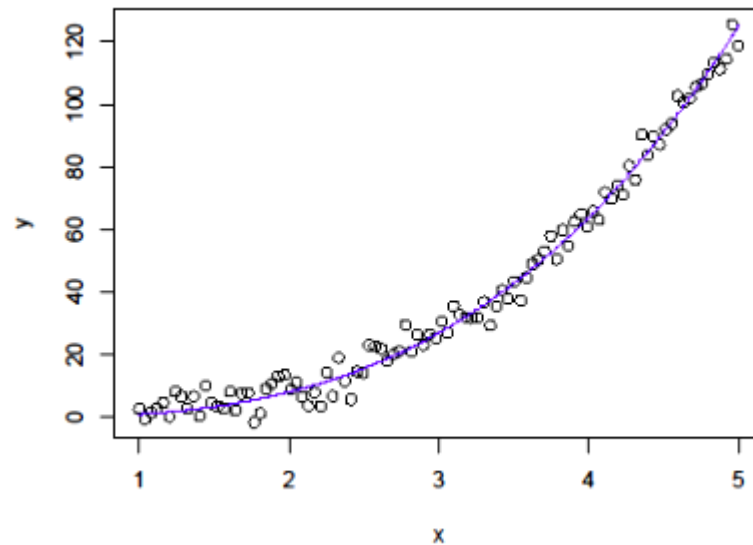
To add a *categorical variable*, use a different plot color or symbol for each category.



Nonlinear Relationships

12

- There are other **forms** of relationships besides linear. The scatterplot below is an example of a **nonlinear form**.
- Note the curvature in the relationship between x and y .



2.3 Correlation

- Correlation coefficient r
- Properties of r

Measuring Linear Association 1

A scatterplot displays the strength, direction, and form of the relationship between **two quantitative variables**. **Linear** relationships are important because a straight line is a simple pattern that is quite common.

Our eyes are not always good judges of how strong a relationship is. Therefore, we use a numerical measure to supplement our scatterplot and help us quantify the strength of the linear relationship.

The **correlation r** measures the **strength** of the **linear** relationship between two **quantitative** variables. Using the notation explained on pp. 103–104 in the text:

$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

Measuring Linear Association 2

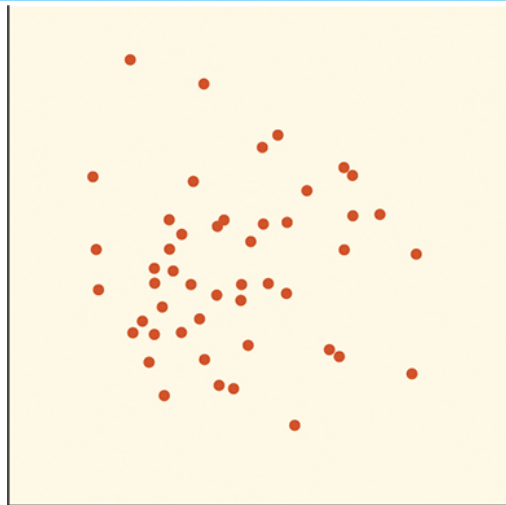
We say a **linear** relationship is **strong** if the points lie close to a straight line and **weak** if they are widely scattered about a line. The following facts about r help us further interpret the strength of the linear relationship.

Properties of Correlation

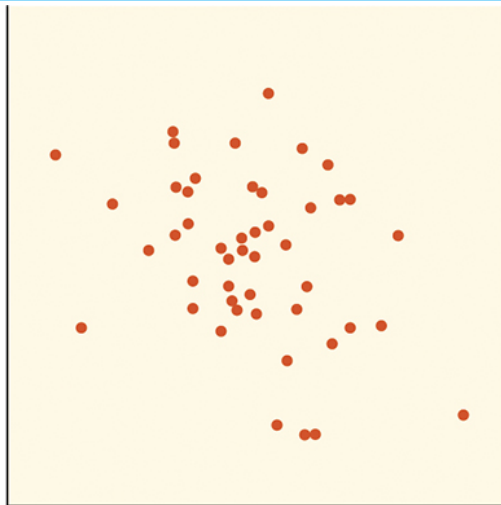
- r is always a number between -1 and 1 .
- $r > 0$ indicates a positive association.
- $r < 0$ indicates a negative association.
- Values of r near 0 indicate a very weak linear relationship.
- The strength of the linear relationship increases as r moves away from 0 toward -1 or 1 .
- The extreme values $r = -1$ and $r = 1$ occur only in the case of a perfect linear relationship.

Correlation

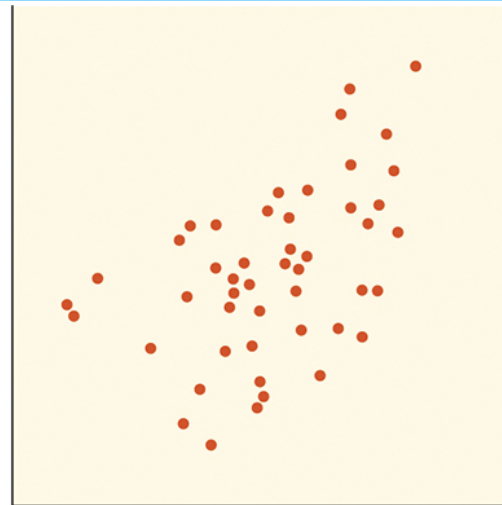
16



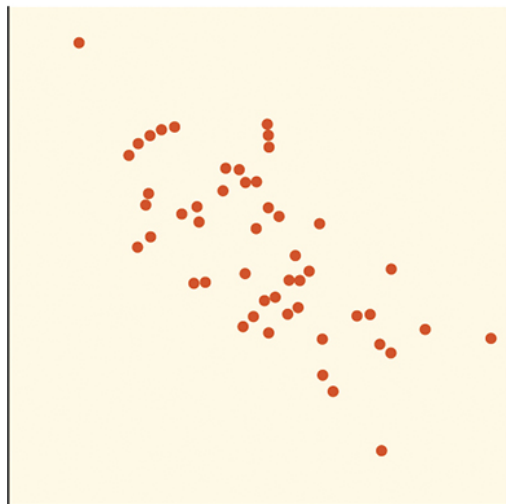
Correlation $r = 0$



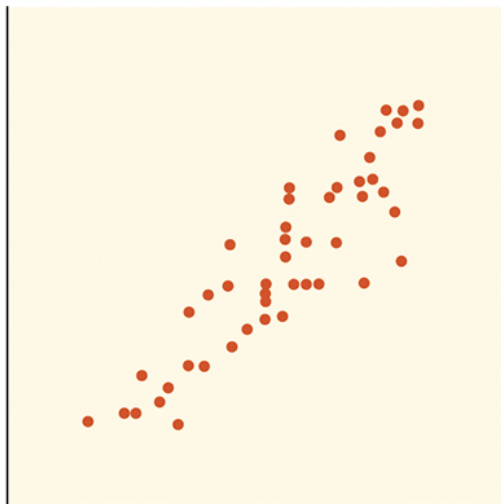
Correlation $r = -0.3$



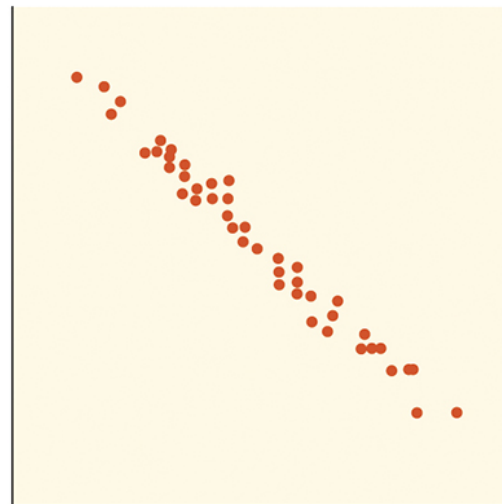
Correlation $r = 0.5$



Correlation $r = -0.7$



Correlation $r = 0.9$



Correlation $r = -0.99$

Properties of Correlation

17

1. Correlation makes no distinction between explanatory and response variables.
2. r has *no units* and *does not change when we change the units of measurement of x , y , or both (why?)*.
3. Positive r indicates positive association between the variables, and negative r indicates negative association.
4. The correlation r is always a number between -1 and 1 .

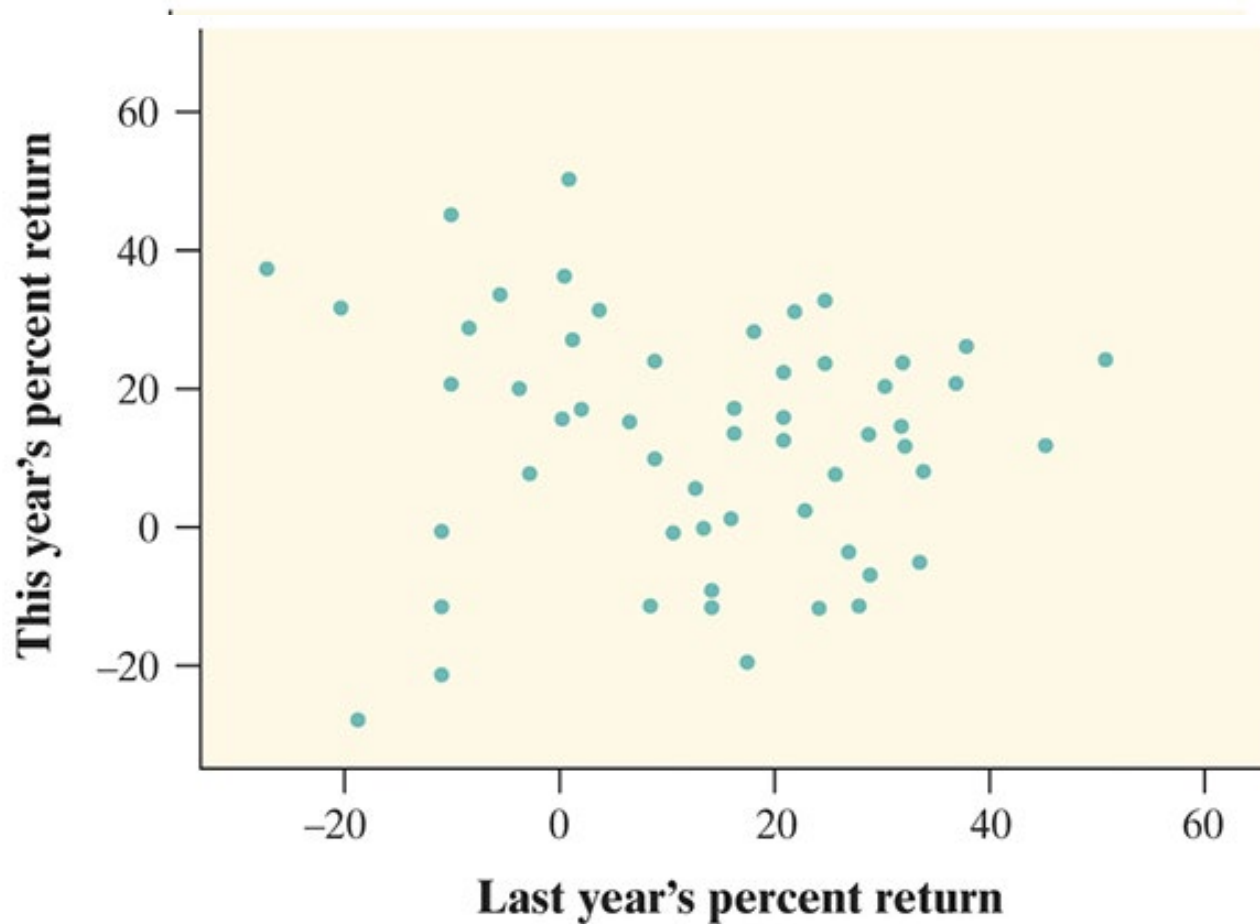
Cautions:

- Correlation requires that both *variables be quantitative*.
- Correlation *does not describe curved relationships* between variables, no matter how strong the relationship is.
- The correlation r is not resistant; it can be strongly affected by a few outlying observations.
- *Correlation is not a complete summary of two-variable data.*

Correlation Example

18

Estimate the correlation r and interpret it in context.



(d)

2.4 Least-Squares Regression

- Regression lines
- Least-squares regression line
- Predictions
- Facts about least-squares regression
- Correlation and regression

Regression Line 1

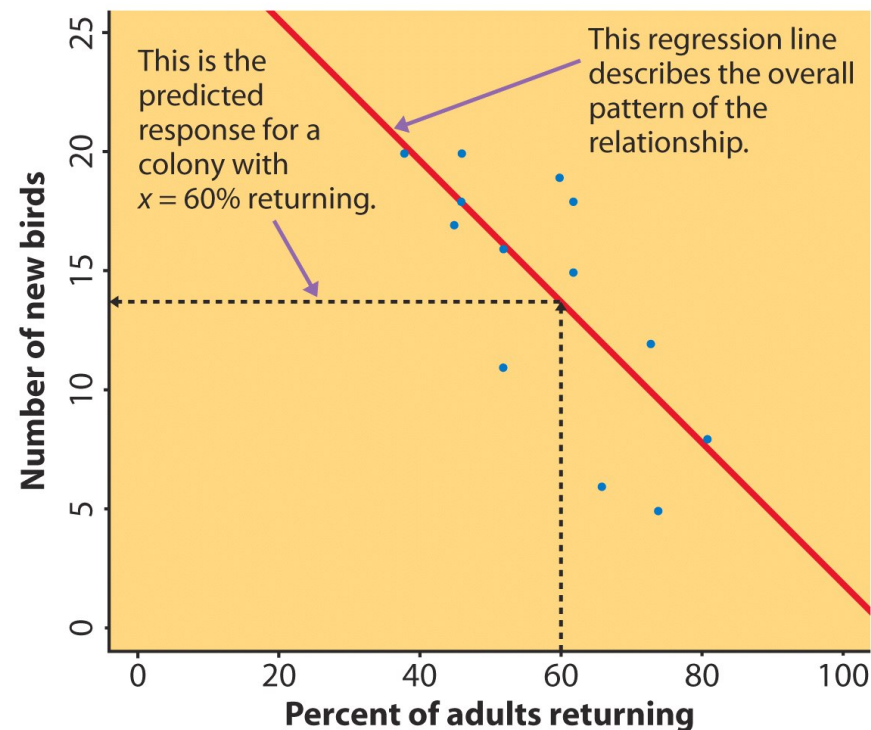
20

A **regression line** is a straight line that describes **how a response variable y changes as an explanatory variable x changes**.

We can use a regression line to predict the value of y for a given value of x .

Example: Predict the number of new adult birds that join the colony based on the percent of adult birds that return to the colony from the previous year.

- If 60% of adults return, how many new birds are predicted?



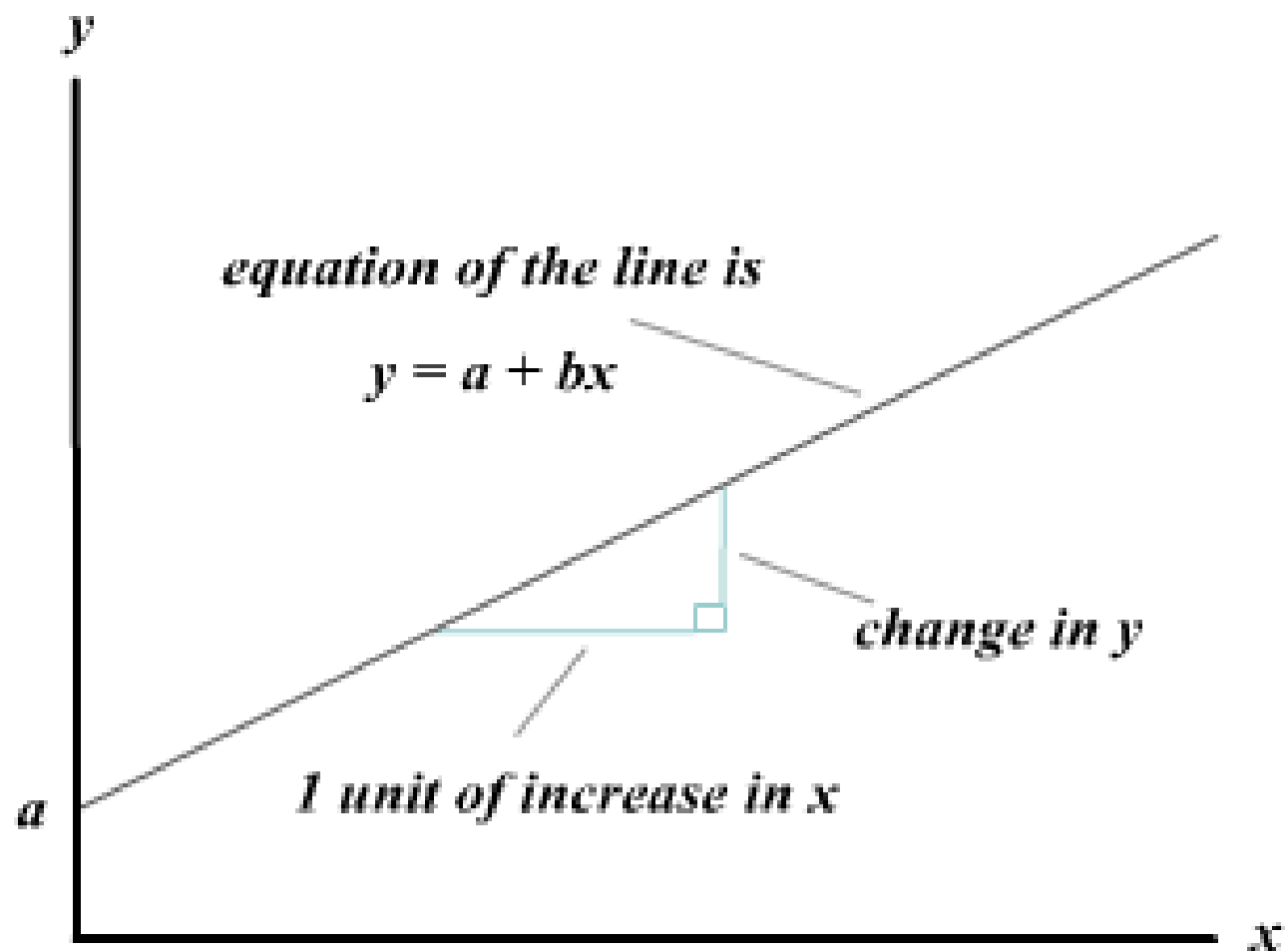
Regression Line 2

When a scatterplot displays **a linear pattern**, we can describe the overall pattern by drawing a straight line through the points. **Fitting a line** to data means drawing a line that comes as close as possible to the points.

Regression equation:

$$\hat{y} = b_0 + b_1x$$

- **x** is the value of the explanatory variable.
- **“y-hat”** is the predicted value of the response variable for a given value of x.
- **b₁** is the **slope**, the amount by which the predicted value of y changes for each one-unit increase in x.
- **b₀** is the **intercept**, the predicted value of y when x = 0.



Least-Squares Regression Line

23

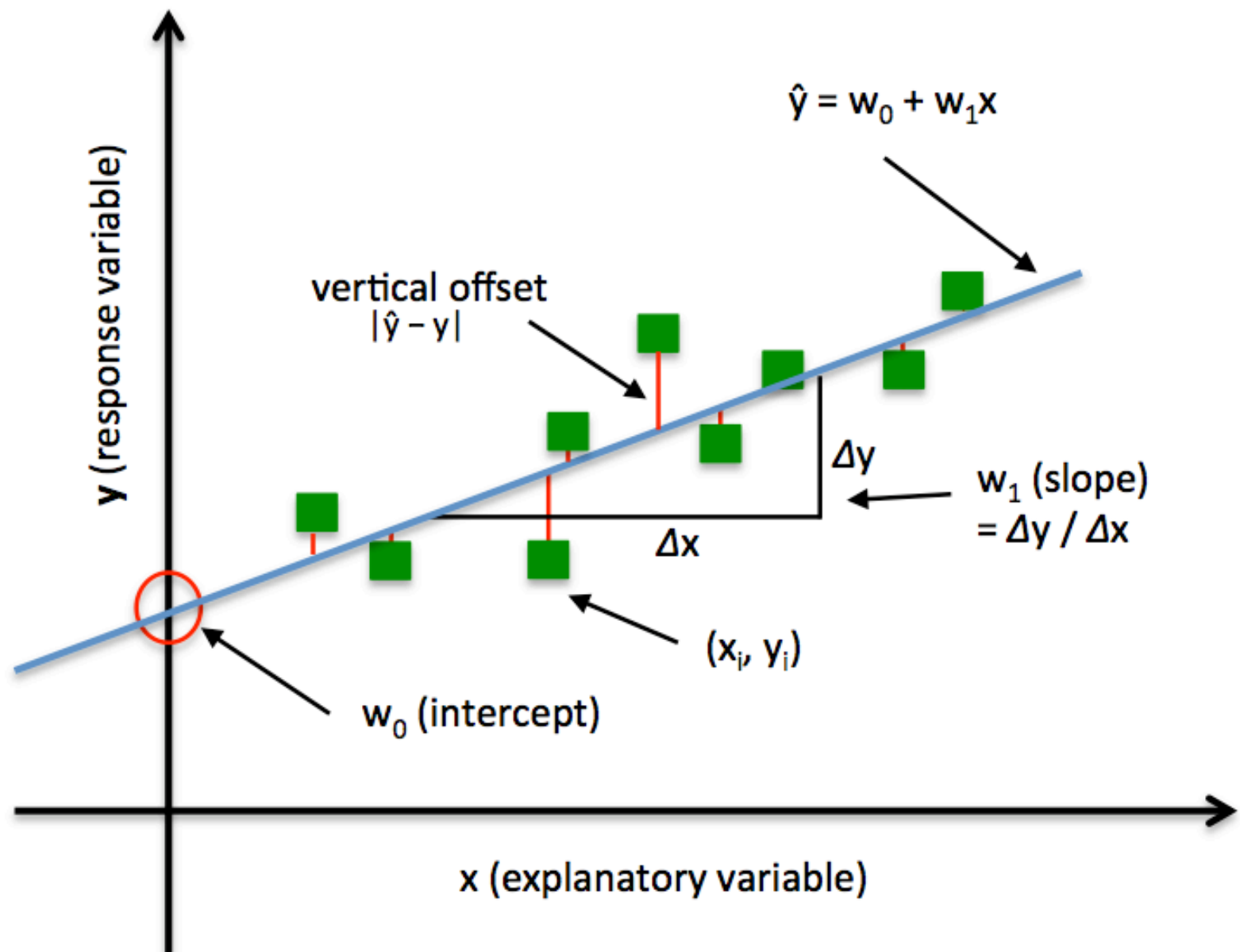
Because we are trying to predict y , we want the regression line to be as close as possible to the data points in **the vertical (y) direction**.

Least-Squares Regression Line (LSRL):

The **least-squares regression line of y on x** is the line that **minimizes the sum of the squares of the vertical distances** of the data points from the line.

If we have data on an explanatory variable x and a response variable y , the equation of the LSRL is

$$y = b_0 + b_1x$$



Predictions Via Regression Line

25

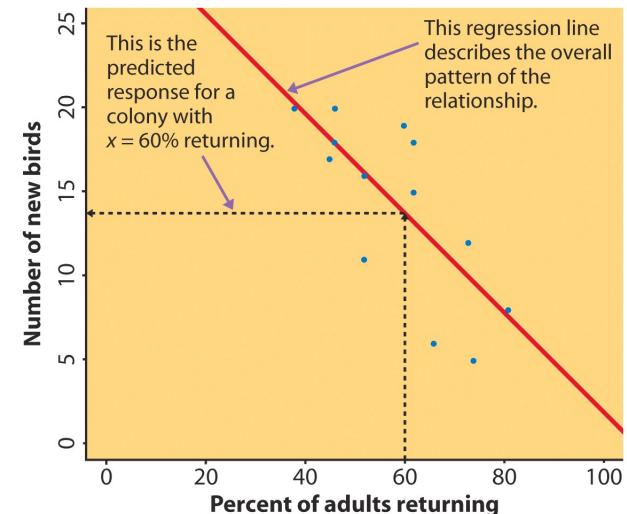
For the returning birds example, the LSRL is
 $\hat{y} = 31.9343 - 0.3040x$

$\hat{y}$ is the **predicted** number of new birds for colonies with x percent of adults returning.

Suppose we know that an individual colony has 60% returning. What would we **predict** the number of new birds to be for just that colony?

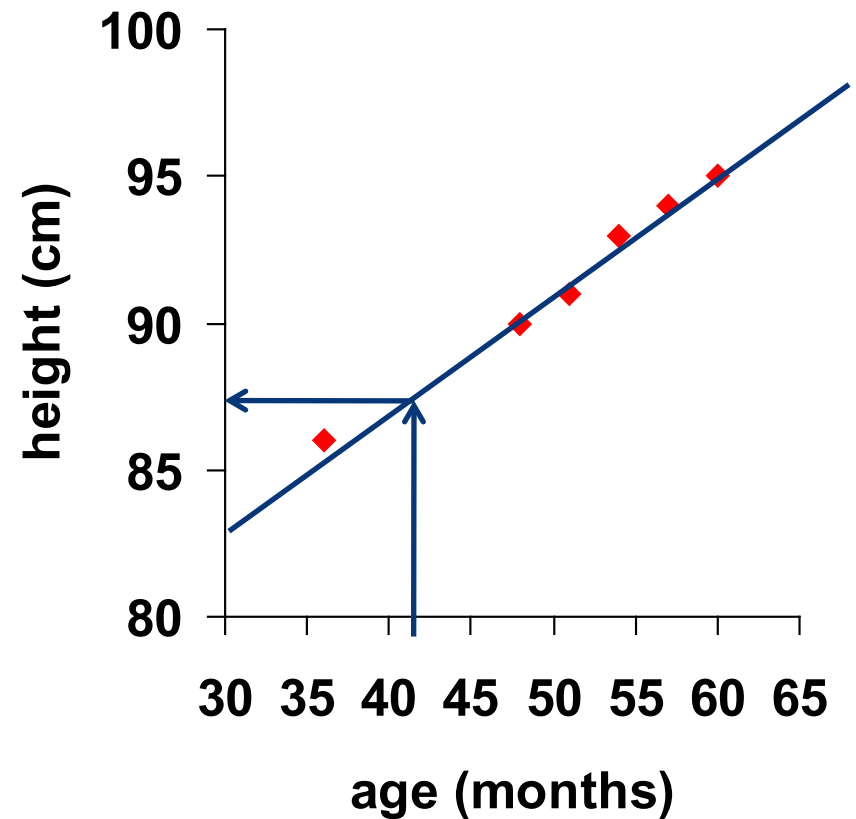
For colonies with 60% returning, we **predict** the **average** number of new birds to be

$$31.9343 - (0.3040)(60) = 13.69 \text{ birds}$$



Extrapolation 1

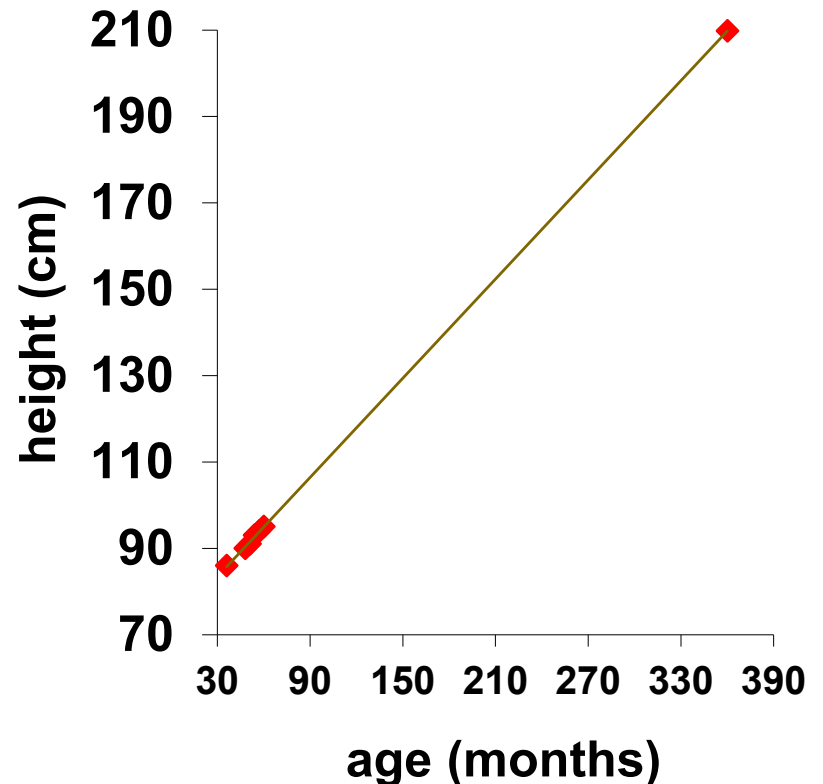
- Sarah's height was plotted against her age.
- Can you guess (predict) her height at age 42 months?
- Can you predict her height at age 30 years (360 months)?



Extrapolation 2

27

- Regression line:
 $\hat{y} = 71.95 + .383 x$
- Height at age 42 months?
 $\hat{y} = 88$
- Height at age 30 years?
 $\hat{y} = 209.8$
- She is predicted to be 6' 10.5" at age 30! *What's wrong?*



Facts About Least-Squares Regression

Regression is one of the most common statistical settings, and least-squares is the most common method for fitting a regression line to data. Here are some facts about least-squares regression lines.

- **Fact 1:** A change of one standard deviation in x corresponds to a change of r standard deviations in y . $b_1 = r \frac{s_y}{s_x}$
- **Fact 2:** The LSRL always passes through ($\bar{x}$, $\bar{y}$)
- **Fact 3:** The distinction between explanatory and response variables is essential.

Correlation and Regression

Least-squares regression looks at the distances of the data points from the line **only in the y direction**. As a result, the variables x and y play different roles in regression. Even though correlation r ignores the distinction between x and y , there is a close connection between correlation and regression.

The **square of the correlation, r^2 or R^2** , (pronounced “*R squared*”) is the fraction of the variation in values of y that is explained by the least-squares regression of y on x .

- r^2 is sometimes called the **coefficient of determination**.

$$b_1 = r \frac{s_y}{s_x}$$

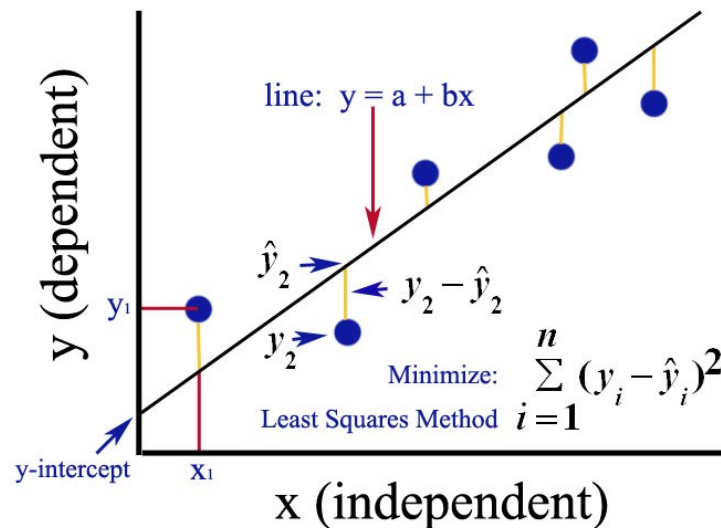
2.5 Cautions about Correlation and Regression

- Residuals and residual plots
- Outliers and influential observations
- Lurking variables
- Correlation and causation

A **residual** is the signed difference between an observed value of response and its corresponding predicted value:

$$\text{residual} = \text{observed } y - \text{predicted } y$$

$$= y - \hat{y}$$



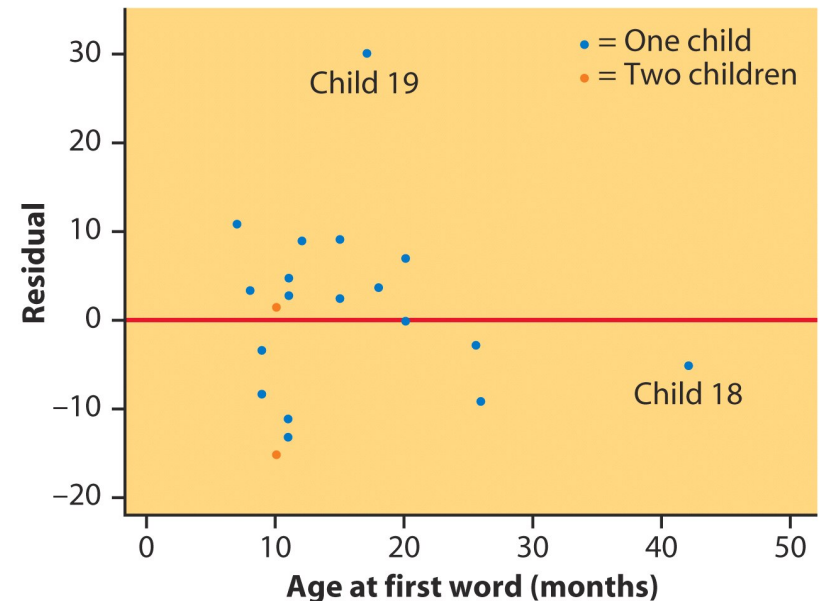
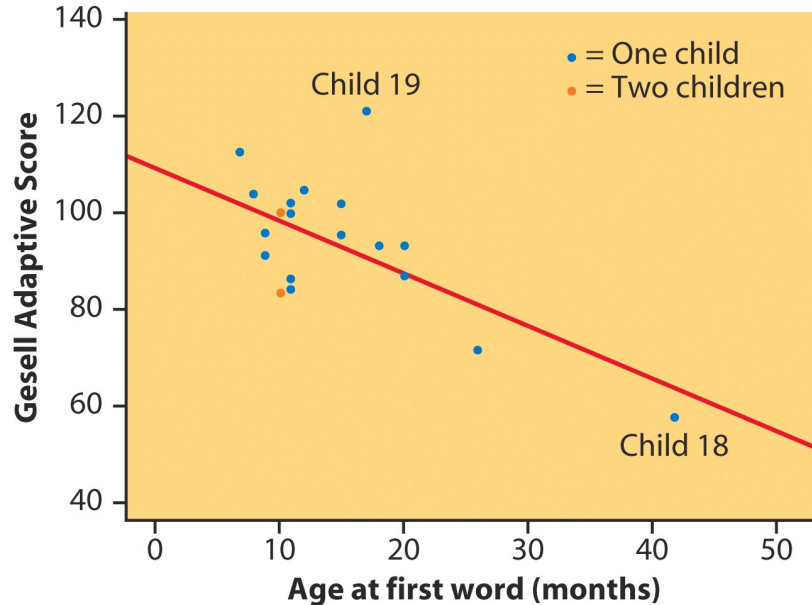
Residual Plots

32

A **residual plot** is a scatterplot of the regression residuals against the explanatory variable. Residual plots help us **assess the fit** of a regression line.

- Ideally there should be a “**random**” scatter around zero.
- Residual *patterns* suggest deviations from a **linear** relationship.

Gesell Adaptive Score and Age at First Word

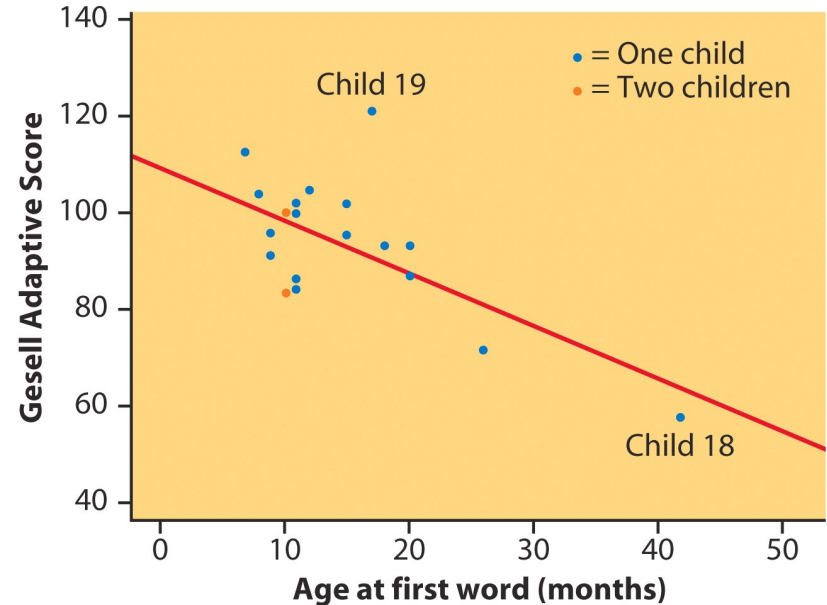


Outliers and Influential Points 1

An **outlier** is an observation that lies outside the overall pattern of the other observations.

- Outliers in the **y** direction have large residuals.
- Outliers in the **x** direction are often **influential** observations for the regression line.

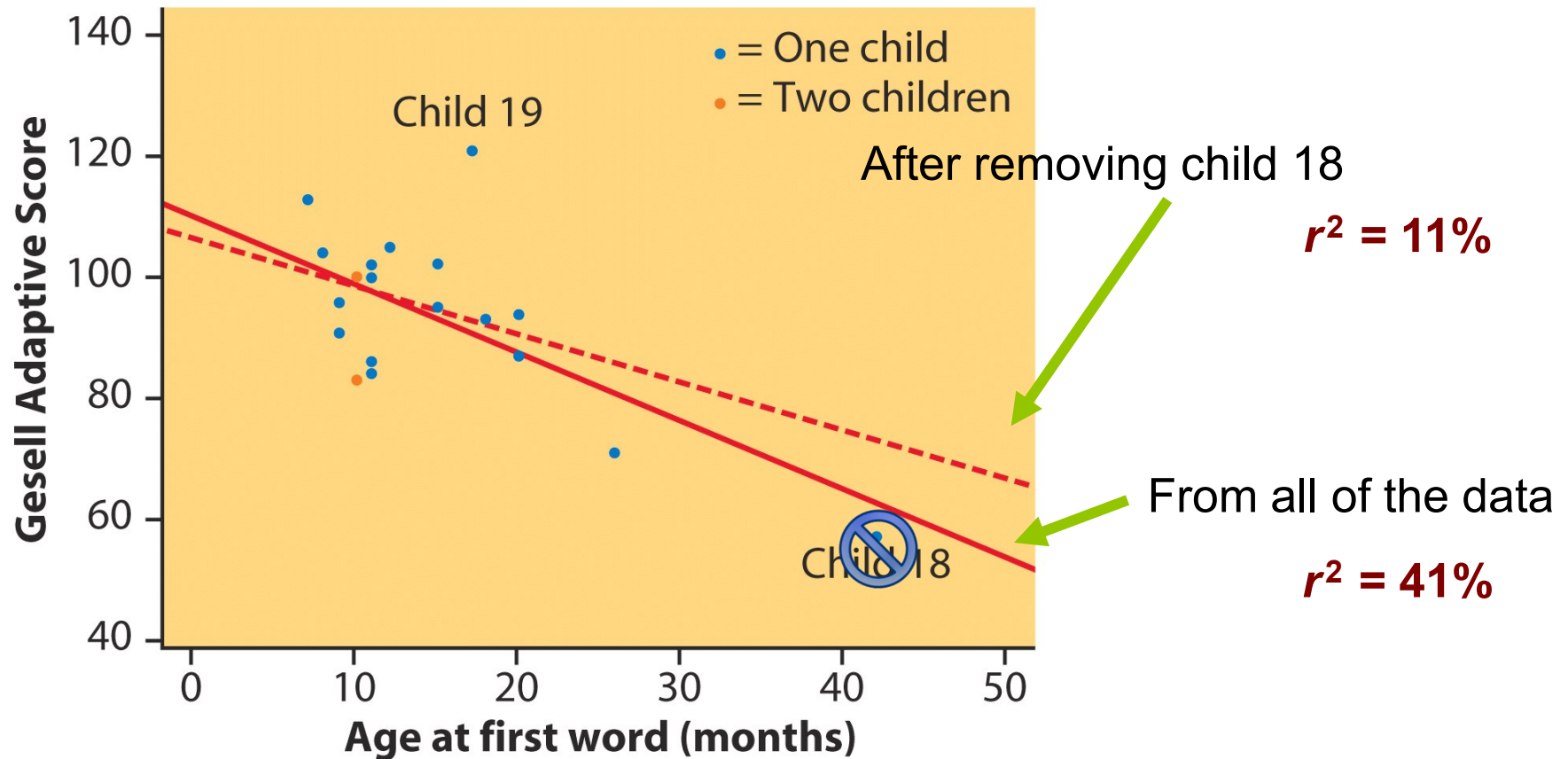
An **influential** observation is one whose removal substantially changes the fitted regression.



Outliers and Influential Points 2

34

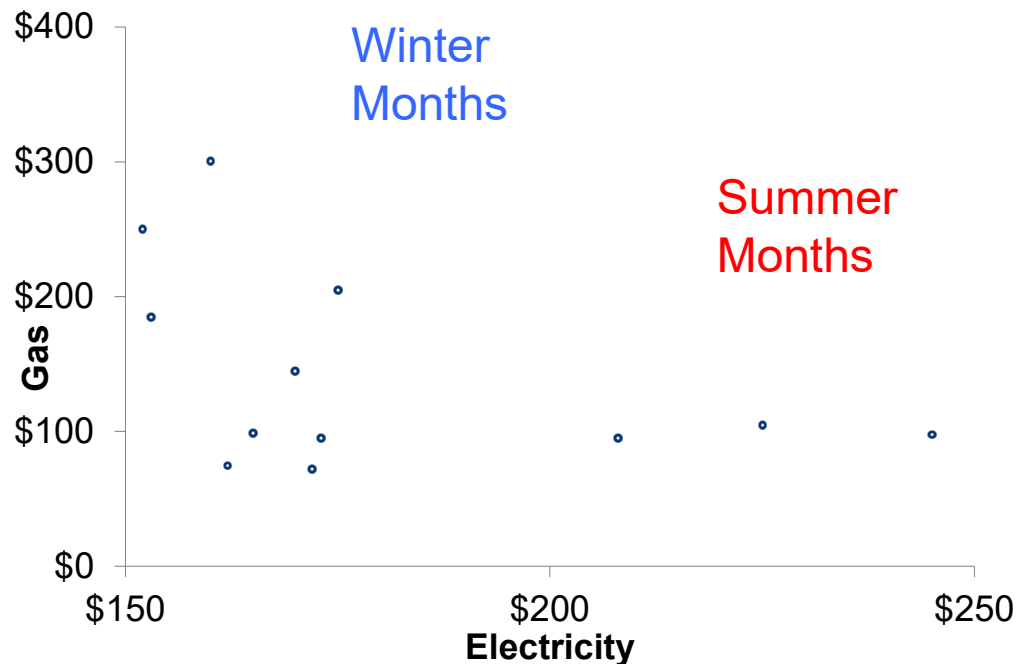
Gesell Adaptive Score and Age at First Word



Lurking Variables

35

A **lurking variable** is a variable that is not among the explanatory or response variables and yet may influence the interpretation of relationships among those variables.



Gas expense seems to be negatively associated with Electricity expense, but “season” is a lurking variable.

Cautions About Correlation and Linear Regression

36

- Both correlation and linear regression describe the linear relationship.
- Both can be strongly affected by outliers/influential observations.
- **Always plot the data before interpreting.**
- Beware of ***extrapolation***.
 - Use caution in predicting y when x is outside the range of observed x 's.
- Beware of ***lurking variables***.
 - A lurking variable is a variable that is not among the explanatory or response variables in a study but may influence the relationship between them.
- **Correlation does not imply causation!**

2.6 Data Analysis for Two-Way Tables

- The two-way table
- The Two-Way Table
- Marginal distributions
- Conditional distributions
- Simpson's paradox

Categorical Variables

38

Recall that categorical variables place individuals into one of several groups or categories.

- The values of a categorical variable are labels for the different categories (i.e, male and female for “sex” variable).
- The distribution of a categorical variable lists the **count** or **percent** of individuals who fall into each category.

When a dataset involves two categorical variables, we may create a two-way table.

A **two-way table** describes two categorical variables, organizing counts according to a **row variable** and a **column variable**. Each combination of values for these two variables is called a **cell**.

The Two-Way Table

39

Young adults by gender and chance of getting rich by age 30

	Female	Male	Total
Almost no chance	96	98	194
Some chance, but probably not	426	286	712
A 50–50 chance	696	720	1416
A good chance	663	758	1421
Almost certain	486	597	1083
Total	2367	2459	4826

What are the variables described by this two-way table?

How many young adults were surveyed?

Marginal Distribution 1

40

The **marginal distribution** of one of the categorical variables in a two-way table is the distribution of that variable **ignoring the other variable**.

Note: Percentages are often more informative than counts, especially when comparing groups of different sizes.

To examine a marginal distribution:

1. Use the data in the table to calculate the marginal distribution (in percent) of the row or column totals.
2. Make a graph to display the marginal distribution.

Marginal Distribution 2

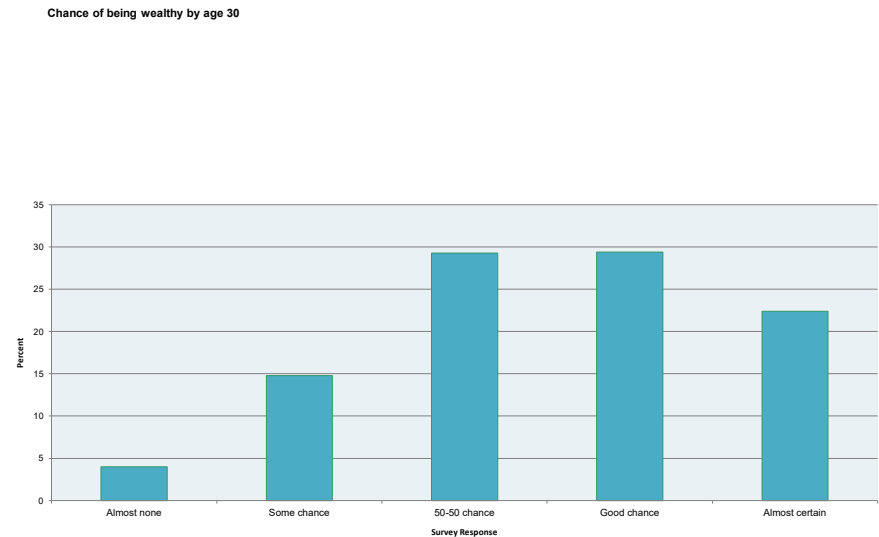
41

Young adults by gender and chance of getting rich

	Female	Male	Total
Almost no chance	96	98	194
Some chance, but probably not	426	286	712
A 50–50 chance	696	720	1416
A good chance	663	758	1421
Almost certain	486	597	1083
Total	2367	2459	4826

Examine the **marginal distribution** of **chance of getting rich**.

Response	Percent
Almost no chance	$194/4826 = 4.0\%$
Some chance	$712/4826 = 14.8\%$
A 50–50 chance	$1416/4826 = 29.3\%$
A good chance	$1421/4826 = 29.4\%$
Almost certain	$1083/4826 = 22.4\%$



Conditional Distribution 1

42

Marginal distributions tell us nothing about the relationship between two variables. For that, we need to explore the conditional distributions of the variables.

A **conditional distribution** of a variable describes the distribution of that variable among individuals who **have a specific value of the other variable**.

To examine or compare conditional distributions:

1. Select the row(s) or column(s) of interest.
2. Use the data in the table to calculate the conditional distribution (**in percent**) of the row(s) or column(s).
3. Make a graph to display the conditional distribution.
 - Use a **side-by-side bar graph** or **segmented bar graph** to compare distributions.

Conditional Distribution 2

43

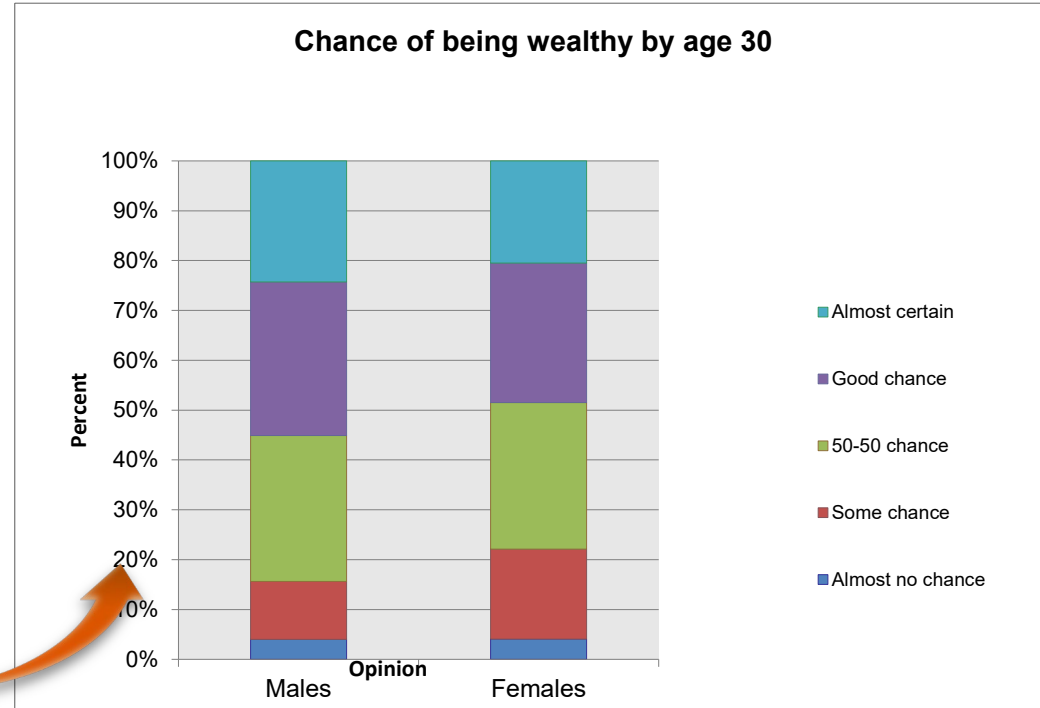
Young adults by gender and opinion of getting rich

	Female	Male	Total
Almost no chance	96	98	194
Some chance, but probably not	426	286	712
A 50–50 chance	696	720	1416
A good chance	663	758	1421
Almost certain	486	597	1083
Total	2367	2459	4826

Calculate the conditional distribution of opinion among males.

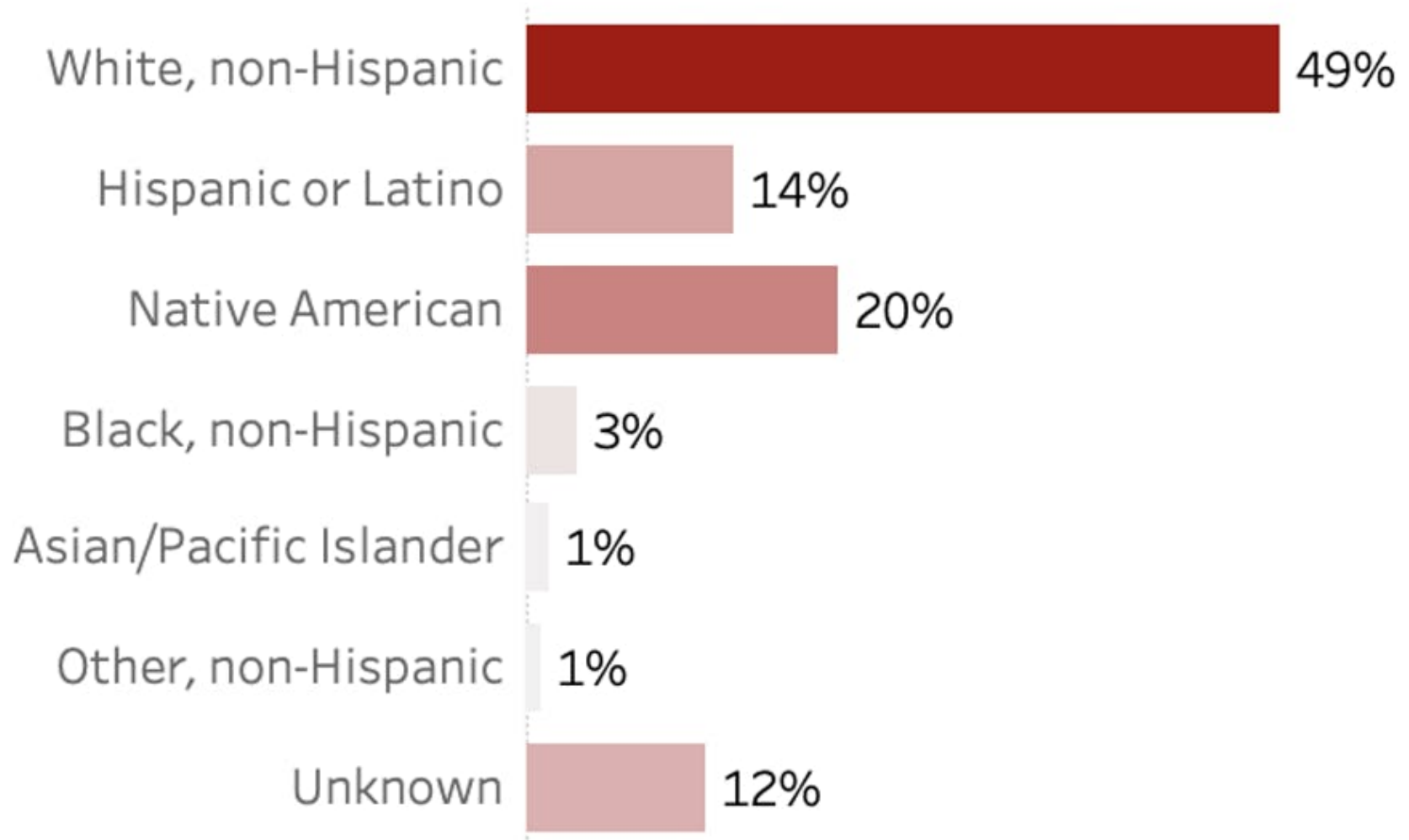
Examine the relationship between gender and opinion.

Response	Male	Female
Almost no chance	$98/2459 = 4.0\%$	$96/2367 = 4.1\%$
Some chance	$286/2459 = 11.6\%$	$426/2367 = 18.0\%$
A 50–50 chance	$720/2459 = 29.3\%$	$696/2367 = 29.4\%$
A good chance	$758/2459 = 30.8\%$	$663/2367 = 28.0\%$
Almost certain	$597/2459 = 24.3\%$	$486/2367 = 20.5\%$

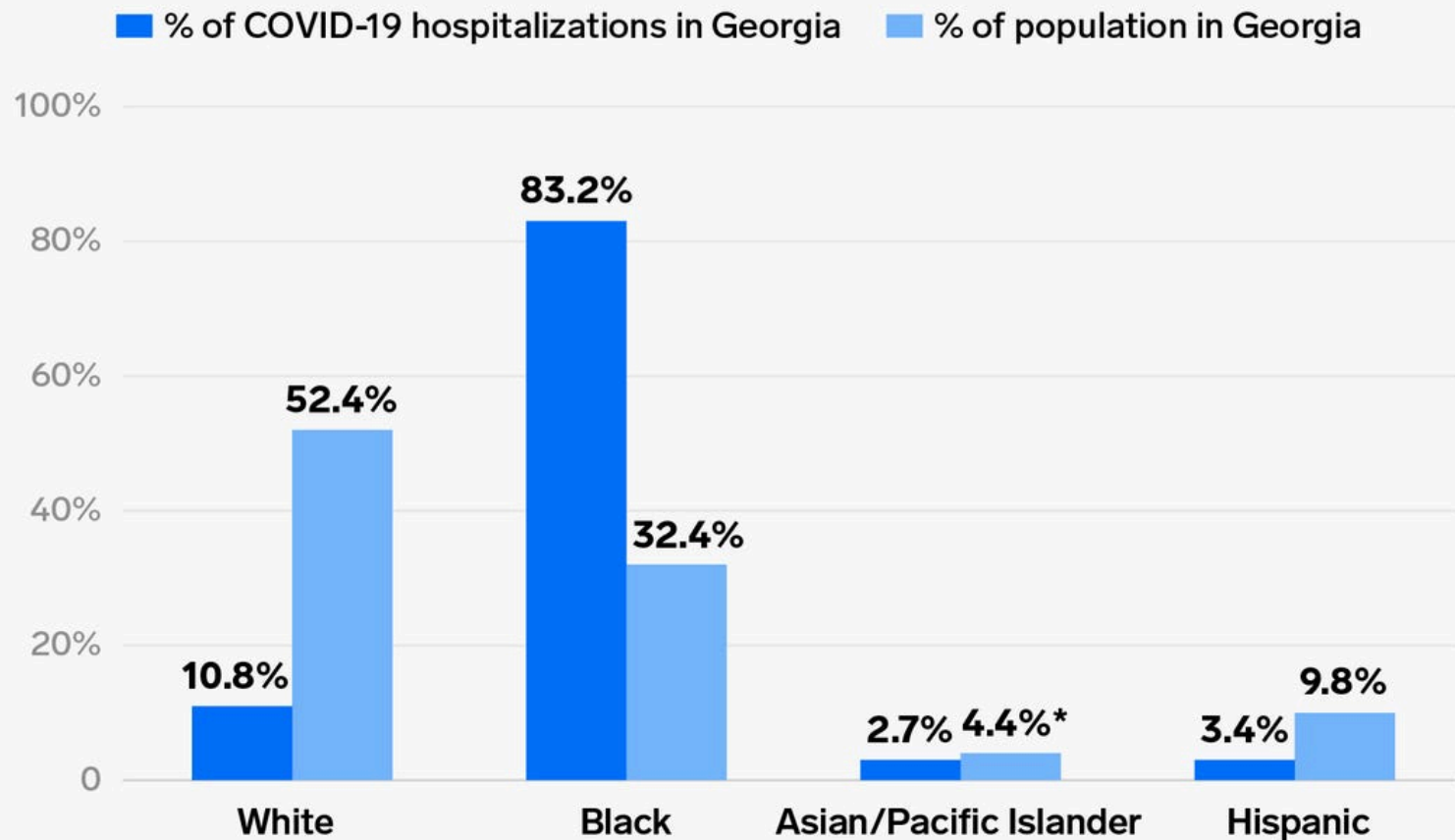




COVID-19 Deaths by Race/Ethnicity



Georgia state COVID-19 hospitalizations, by race



*Combined US census results for "Asian alone" and "Native Hawaiian and Other Pacific Islander alone."

Note: Data reflects March hospitalizations across 8 Georgia hospitals. Data on race/ethnicity was known for 297 of 305 patients.

Sources: CDC; US Census Bureau

INSIDER



Simpson's Paradox 1

46

An association or comparison that holds for all or several groups can reverse direction when the data are combined to form a single group. This reversal is called **Simpson's paradox**.

Potential cause: When studying the relationship between two variables, there might exist a **lurking variable** that creates a reversal in the direction of the relationship when the lurking variable is ignored as opposed to the direction of the relationship when the lurking variable is considered.

The lurking variable creates subgroups, and failure to take these subgroups into consideration can lead to misleading conclusions regarding the association between the two variables.

Simpson's Paradox 2

47

Consider the acceptance rates for the following groups of men and women who applied to college.

Counts	Accepted	Not accepted	Total
Men	198	162	360
Women	88	112	200
Total	286	274	560

Percents	Accepted	Not accepted
Men	55%	45%
Women	44%	56%

A higher percentage of men were accepted: Is there evidence of discrimination?

Simpson's Paradox 3

48

Consider the acceptance rates when broken down by type of school.

BUSINESS SCHOOL

Counts	Accepted	Not accepted	Total
Men	18	102	120
Women	24	96	120
Total	42	198	240

Percents	Accepted	Not accepted
Men	15%	85%
Women	20%	80%

ART SCHOOL

Counts	Accepted	Not accepted	Total
Men	180	60	240
Women	64	16	80
Total	244	76	320

Percents	Accepted	Not accepted
Men	75%	25%
Women	80%	20%

Simpson's Paradox 4

49

- ✓ **Lurking variable:** Applications were split between the Business School (240) and the Art School (320).

Within each school a higher percentage of women were accepted than men.

There is not any discrimination against women!

This is an example of **Simpson's paradox**.

When the lurking variable (Type of School: Business or Art) is ignored, the data seem to suggest discrimination against women.

However, when the type of school is considered, the association is reversed and suggests discrimination against men.

2.7 The Question of Causation

- Explaining association
- Causation
- Common response
- Confounding
- Establishing causation

Explaining Association: Causation

51

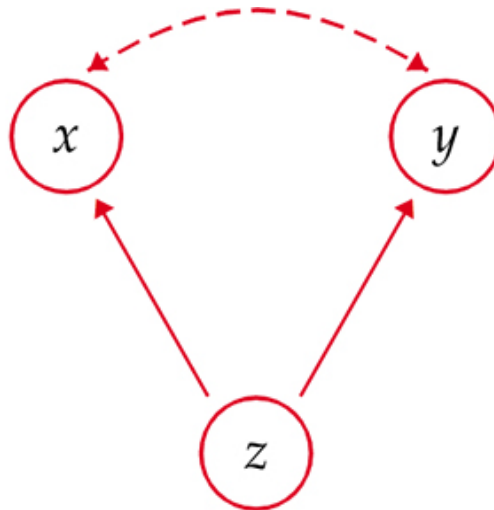
Association, however strong, does NOT imply causation.

Some possible explanations for an observed association

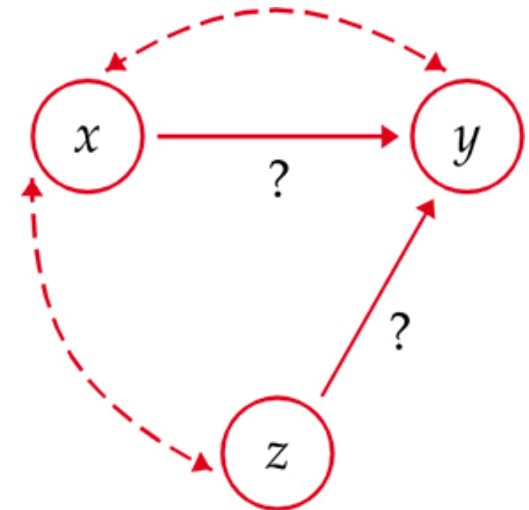
The dashed lines show an association. The solid arrows show a cause-and-effect link. x is explanatory, y is response, and z is a lurking variable.



Causation



Common response



Confounding

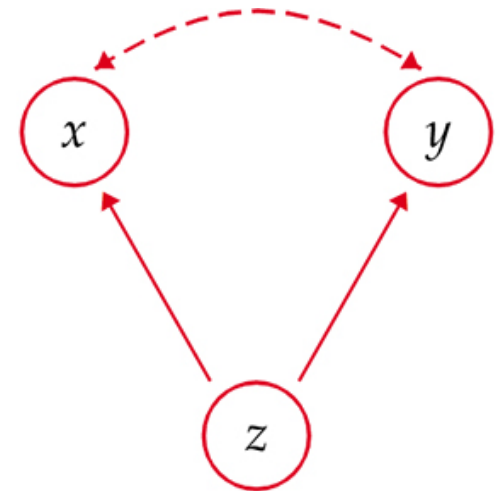
Explaining Association: Common Response

52

“Beware the lurking variable” is good advice when thinking about an association between two variables. The observed relationship between the variables can be explained by a lurking variable. Both x and y may change in response to changes in z .

Most students who have high SAT scores (x) in high school have high GPAs (y) in their first year of college.

- This positive correlation can be explained as a *common response* to students' ability and knowledge.
- The observed association between x and y could be explained by a third lurking variable z . In this example, “ability and knowledge” is the lurking variable.
- Both x and y change in response to changes in z . This creates an association even though there is no direct causal link.



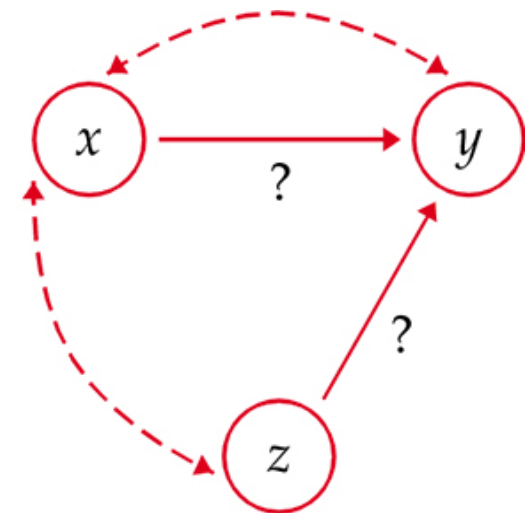
Common response

Explaining Association: Confounding

53

Two variables are **confounded** when their effects on a response variable cannot be distinguished from each other. The confounded variables may be either explanatory variables or lurking variables.

- Example: Studies have found that religious people (z) live longer (y) than nonreligious people.
- Religious people also take better care of themselves and are less likely to smoke (x) or be overweight.



Confounding

Establishing Causation

54

It appears that lung cancer is associated with smoking.

How do we know that both of these variables are not being affected by an unobserved third (lurking) variable?

For instance, what if there is a **genetic predisposition** that causes people to both get lung cancer *and* become addicted to smoking, but the smoking itself does not CAUSE lung cancer?

We can evaluate the association using the following criteria:

1. The association is strong.
2. The association is consistent.
3. Higher doses are associated with stronger responses.
4. Alleged cause precedes the effect.
5. The alleged cause is plausible.

Evidence of Causation

A properly conducted **experiment** may establish causation.

Other considerations when we cannot do an experiment:

- The association is *strong*.
- The association is *consistent*.
 - The connection happens in *repeated trials*.
 - The connection happens under *varying conditions*.
- Higher doses are associated with stronger responses.
- Alleged cause *precedes* the effect in time.
- Alleged cause is *plausible* (reasonable explanation).



Chapter 2

Looking at Data— Relationships (Review)

2.1 Relationships

2.2 Scatterplots

2.3 Correlation

2.4 Least-Squares Regression

**2.5 Cautions about Correlation and
Regression**

2.6 Data Analysis for Two-Way Tables

2.7 The Question of Causation